



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

BUILDING CONSTRUCTION BASIC WELDING

BDCBW401

Apply Construction basic welding

Competence

RQF Level: 4

Learning Hours



30

Credits: 3

Sector: Construction and Building Services

Trade: Building Construction

Module Type: General

Curriculum: CBSBDC4001- TVET Certificate 4In Building Construction

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Purpose statement	This module describes the skills, knowledge and attitude to perform the basics welding on construction sites. It is intended to be pursued by the learner at TVET level IV in Building Construction. At the end of this module, learner will be able to prepare the work by selecting suitable materials, tools and equipment; perform arc welding; and perform oxy-acetylene gas welding as required.						
Leaning assumed to be in place	Safety, health and environment at workplace; Building construction drawing						
Delivery modality	Training delivery			100%	Assessment		Total 100%
	Theoretical content			30%	Formative assessment	30%	50%
	Practical work:			70%		70%	
	• Group project and presentation	30%					
	• Individual project /Work	70%					
				Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria	
1. Prepare the work	1.1.	PPE are properly selected and worn as per safety measures
	1.2.	Safety precautions of working area are appropriately applied before the work as per standard
	1.3.	Workplace is properly cleaned and arranged according to the standard
	1.4.	Tools, equipment and materials are properly selected according to the work to be done
2. Perform arc welding	2.1.	Measuring, marking, and cutting tools are Correctly used according to their types
	2.2.	Edges are correctly prepared according to the standards
	2.3.	Arc welding techniques are correctly applied in all positions as per standards
3. Perform oxyacetylene gas	3.1.	Equipment are properly set according to the standards

welding	3.2. Welding edges are properly prepared according to standards
	3.3. Oxy-acetylene gas welding is properly applied according to the standards techniques

Course content

Learning outcomes	At the end of the module the learner will be able to:
	1. Prepare the work 2. Perform arc welding 3. Perform oxyacetylene gas welding
Learning outcome 1: Prepare the work	Learning hours: 5

Indicative content

- **Selection and wearing of PPE**
 - Identification of PPE
 - Selection of PPE
 - Wearing of PPE
- **Application of safety precautions in working area**
 - Work area safety
 - Tool and equipment safety
 - Personal safety
- **Cleaning and arrangement of the workplace**
 - Identification of cleaning types
 - Cleaning of workplace, tools, materials, and equipment
 - Arrangement of workplace layout
 - Arrangement of first aid tool kit
 - Arrangement of fire extinguishers
 - Arrangement of fire blanket
- **Tools, equipment and materials selection**
 - Types of welding tools
 - Types of welding equipment
 - Types of welding materials

Resources required for the learning outcome

Equipment	Black board, Projector, Computer, dust pan and Air compressor
Materials	Soap solution, Solvent, Clothes rugs, white board, spray bottle, soft brushes, scrubber sponge, Broom, Mops, Marker pen, black board and

	Chalks.
Tools	Electric blower, cleaning drill brush, cleaners' corner & edge brush.
Facilitation techniques	• Group discussion • Trainer guided • Brainstorming
Formative assessment methods	• Written assessment • Performance assessment • Matching questions • Multiple choice • Sentence completion • Short questions • Open ended questions

Learning outcome 2: Perform arc welding		Learning hours:15
Indicative content		
• Types of Arc Welding ➤ Shielded Metal Arc Welding ➤ Gas Metal Arc Welding (MIG or MAG Welding) ➤ Flux-Cored Arc Welding, ... • Edge preparations ➤ Edges preparation techniques • Application of arc welding techniques in all positions ➤ Application of different types of joints to produce building elements (windows, doors, trusses) ✓ Butt joint ✓ Corner joint ✓ Edge joint ✓ Lap joint ✓ T-joint ➤ Application of Welding positions ✓ Flat ✓ Horizontal ✓ Vertical ✓ Overhead		
Resources required for the indicative content		
Equipment	Welding machine, drilling machine, Grinding machine, PPE, Projector, Computer	
Materials	Ferrous-metals, Non-Ferrous metals, welding electrodes, sheet metals.	
Tools	Slot pencil, drawing copy, Ruler, Rubber, Scriber, Center punch, Compasses, square,	

	Drawing board, Bench Vice, Hacksaw, Wire brush, chipping hammer
Facilitation techniques	• Demonstration • simulation • Group discussion • Trainer guided • Brainstorming
Formative assessment methods	• Written assessment • Performance assessment • Matching questions • Multiple choice • Sentence completion • Short questions • Open ended questions

Learning outcome 3: Perform oxy acetylene gas welding	Learning hours: 10
Indicative content	
• Setting up of the equipment pressure ➤ Operation pressure ➤ Cutting pressure • Flame ➤ Brazing flame ➤ Welding flame ➤ Cutting flame • Edge preparations ➤ Edges preparation techniques • Application of oxy-acetylene gas welding ➤ Application of different types of joints to produce building elements (windows, doors, trusses) ✓ Butt joint ✓ Corner joint ✓ Edge joint ✓ Lap joint ✓ T-joint ➤ Application of Welding positions ✓ Flat ✓ Horizontal ✓ Vertical ✓ Overhead	
Resources required for the indicative content	

Equipment	-Oxy-acetylene gas welding equipment and accessories, PPE, Projector, Computer
Materials	-Ferrous-metals, Non-Ferrous metals, filler metals, oxygen, and acetylene gas, sheet metals
Tools	-Slot pencil, drawing copy, Ruler, Rubber, Scriber, Center punch, Compasses, T-square, Drawing board, Bench Vice, Hacksaw, Wire brush, chipping hammer.
Facilitation techniques	• Group discussion • Trainer guided • Brainstorming
Formative assessment methods	• Written assessment • Performance assessment • Matching questions • Multiple choice • Sentence completion • Short questions • Open ended questions

References:

1. Workshop processes practices and materials (by Bruce. Black) 2004, Third Edition
2. Introduction to basic manufacturing process and workshop technology (by Rajadar Singh) 2006
3. Welding-Science-and-Technology (by Md. Ibrahim Khan) 2007
4. Mechanical workshop practice (by K.C John) 2010
5. Introduction to basic manufacturing process and workshop technology (by Rajender singh) 2006
6. Production technology (manufacturing processes, Technology and Automation) by R.K. Jain 2012